Grafana Labs

Zero to One: Simplifying Kubernetes Monitoring with Grafana's Helm Chart

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Looks familiar?

The overwhelming landscape of Kubernetes Monitoring

Fragmented Tooling Landscape

Complex Configuration And Setup

No Instant Visibility And Scalability

We deserve better

Single command, instrument everything in minutes

What did we actually install?

Grafana Alloy

*Grafana's open source distribution of the **OpenTelemetry Collector***

What is it?

Alloy is our “Big Tent” collector that pulls together multiple OSS observability signals. It provides our users with an option that is optimized for the LGTM stack, simplifies telemetry connection, blends OTel + Prometheus signals, and supports Grafana Cloud solutions.

Native Kubernetes support

Production-ready dashboards + alerts

Share and reuse pipelines

Centralized configuration support

Programmable pipelines

Grafana Alloy

Clustering support

Built-in pipeline debugging

100+ components

Logs

Metrics

Traces

Profiles

Node Exporter

*Prometheus exporter for hardware and OS metrics on *NIX systems*

What is it?

Node Exporter is a foundational component in the open-source Prometheus ecosystem for infrastructure monitoring. It provides the essential node-level metrics that underpin health and state insights across systems, powering features in community-driven monitoring solutions.

And more...

Automatic Instrumentation
With the Helm chart, you can install a Grafana Beyla DaemonSet to perform eBPF-based zero code instrumentation of applications and gather network metrics.

Kubernetes Metrics
A kube-state-metrics Deployment listens to the Kubernetes API server and generates metrics on the health of objects inside the Cluster such as Deployments, Nodes, and Pods.

Container Metrics
Native integration with cAdvisor handles container CPU, memory and disk usage monitoring.

Energy Metrics
You can opt-in to gather energy metrics powered by the Kepler project.

Cost Metrics
Managing your cluster costs was also made simple by an out-of-the-box integration with OpenCost.

Let’s take a look at our cluster, shall we?

grafana/k8s-monitoring-helm

Instrumentation done, now what?

Long term storage

Grafana Cloud

Your hosted observability stack

You can get started for free. Single pane of glass for all your observability data.

Seamlessly integrate with 100+ data sources. Use prebuilt dashboards, AI anomaly detection, and incident tools to speed troubleshooting.

Scale to billions of metrics with high availability, cost-efficient storage, and enterprise security (SSO, RBAC, auto-updates).

vs

Self Hosted

Mix and match yourself

You can self host all of our open source projects, including Grafana, Mimir, Loki, Tempo, Pyroscope and more.

You can choose to send your data elsewhere too, Grafana Alloy is an OpenTelemetry collector after all. No vendor lock-in.

Blinkuu/talks

Thank you!

Have more questions?

Join us at community.grafana.com

or email me at lukasz.gut@grafana.com

Get involved:

#kubernetes

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